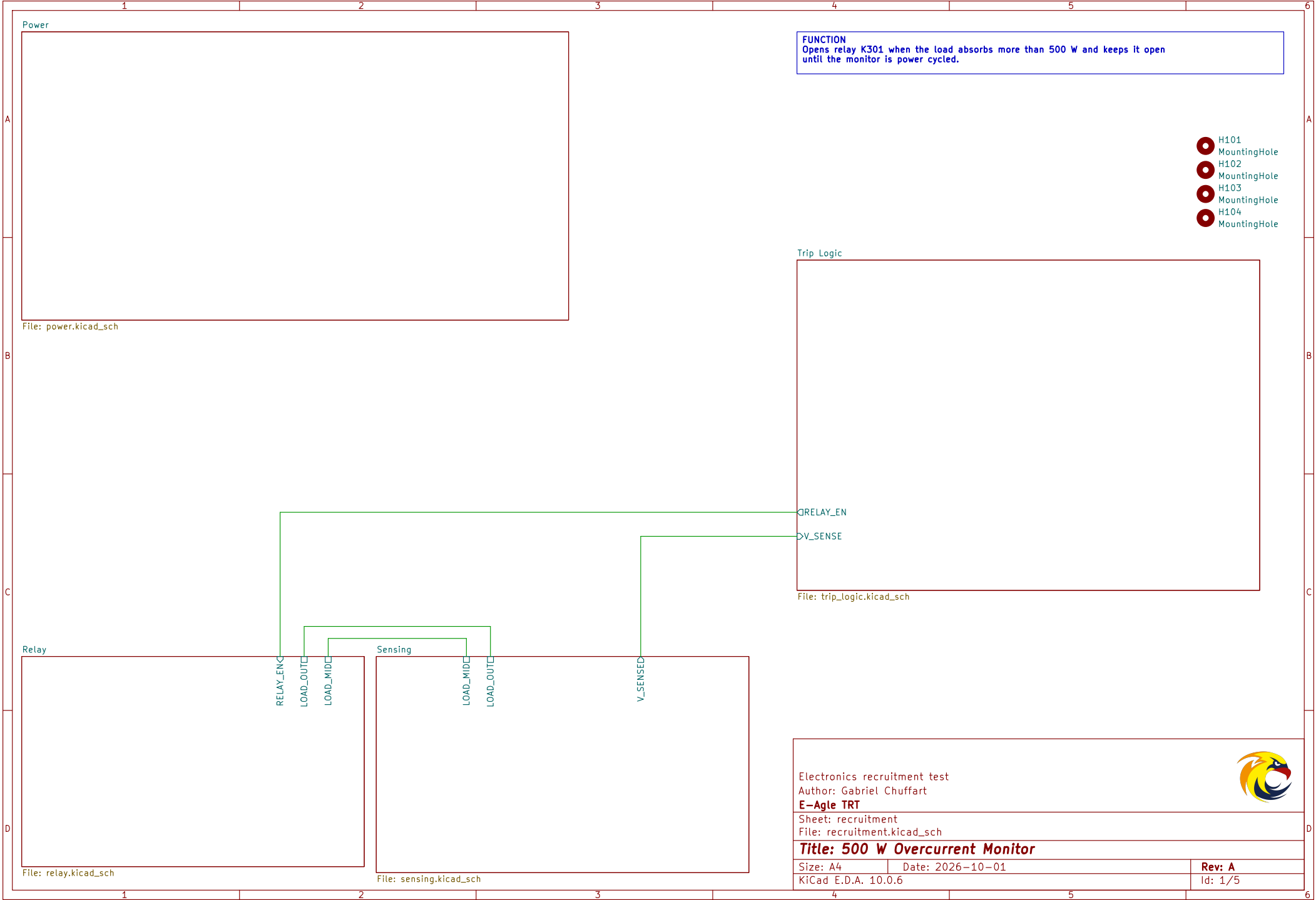




Power

File: power.kicad_sch

FUNCTION
Opens relay K301 when the load absorbs more than 500 W and keeps it open until the monitor is power cycled.

- H101 MountingHole
- H102 MountingHole
- H103 MountingHole
- H104 MountingHole

Trip Logic

File: trip_logic.kicad_sch

Relay

File: relay.kicad_sch

Sensing

File: sensing.kicad_sch

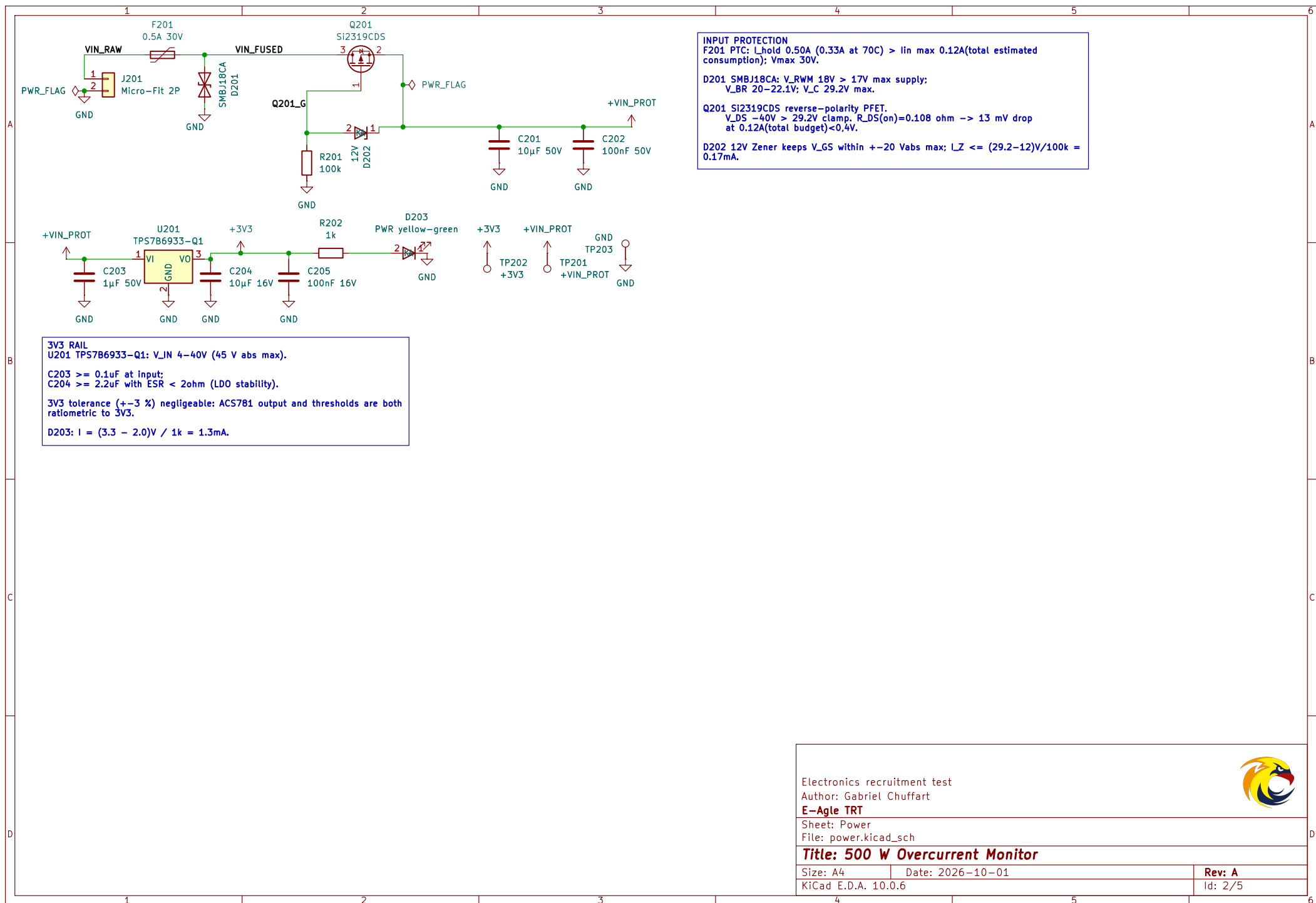
Electronics recruitment test
Author: Gabriel Chuffart
E-Agle TRT



Sheet: recruitment
File: recruitment.kicad_sch

Title: 500 W Overcurrent Monitor

Size: A4	Date: 2026-10-01	Rev: A
KiCad E.D.A. 10.0.6		Id: 1/5



Electronics recruitment test

Author: Gabriel Chuffart

E-Agle TRT

Sheet: Power

File: power.kicad_sch

Title: 500 W Overcurrent Monitor

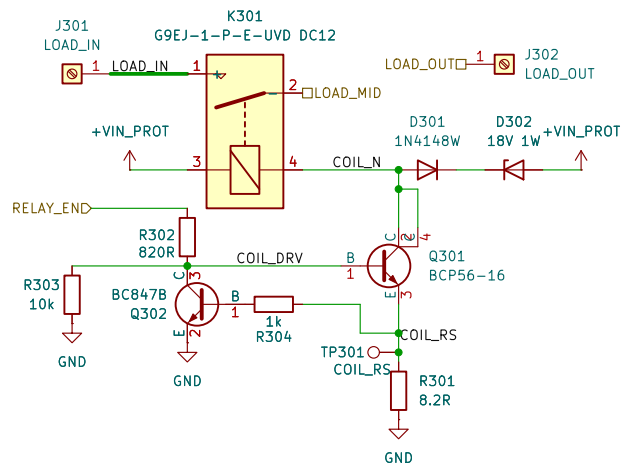
Size: A4 Date: 2026-10-01

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Rev: A

Id: 2/5





COIL DRIVER – constant-current sink: Q301 (BCP56–16), Q302 (BC847B), R301(8.2 ohm)

Holds the coil current at 78mA regardless of the 9–17V supply without exceeding coil rating.

Q302 turns Q301 down as soon as the voltage across R301 reaches its VBE.

Coil current: $I = V_{BE}(Q302) / R301 = 0.64V / 8.2ohm = 78\text{ mA}$

R303 (10k): holds Q301 off if RELAY_EN is unpowered or floating, so the relay stays open.

COIL TURN-OFF PROTECTION – D301 (1N4148W) in series with D302 (18 V, 1 W Zener)

When Q301 switches off, the coil current needs a path to not damage Q301

Omron recommends diode + Zener for this relay (datasheet precaution 14).

Electronics recruitment test
Author: Gabriel Chuffart

E-Agle TRT

Sheet: Relay
File: relay.kicad_sch

Title: 500 W Overcurrent Monitor

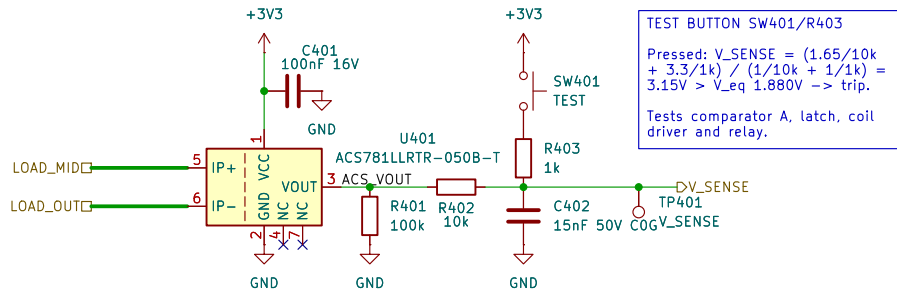
Size: A4 Date: 2026–10–01

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Id: 3/5





CURRENT SENSOR U401 ACS781LLRTR-050B-T
 $V_{OUT} = V_{CC}/2 + S \times I$, $S = 26.4\text{mV/A}$ at $V_{CC} = 3.3\text{V}$.
 Load range $[-3;+25]\text{A} \rightarrow V_{OUT} [1.571;2.310]\text{V}$.
 C401 100nF bypass, cf datasheet.
 R401 100k pull-down: sensor unpowered or VOUT open
 $\rightarrow V_{OUT} = 0\text{V} \rightarrow$ sensor-fault failsafe

INPUT FILTER R402/C402
 $f_c = 1 / (2 \pi \times 10\text{k} \times 15\text{nF}) = 1.06\text{kHz}$, $\tau = 150\text{us}$.
 Sensor noise 36mVpp (1.4App) at 120 kHz bandwidth $\rightarrow$
 $\sim 3.4\text{mVpp}$ (0.13App) after filter.
 R402 isolates C402 from VOUT (cf datasheet: max 10nF
 directly on VOUT).

TEST BUTTON SW401/R403
 Pressed: $V_{SENSE} = (1.65/10\text{k} + 3.3/1\text{k}) / (1/10\text{k} + 1/1\text{k}) = 3.15\text{V} > V_{eq} 1.880\text{V} \rightarrow$ trip.
 Tests comparator A, latch, coil driver and relay.

Electronics recruitment test
 Author: Gabriel Chuffart

E-Agle TRT

Sheet: Sensing
 File: sensing.kicad_sch

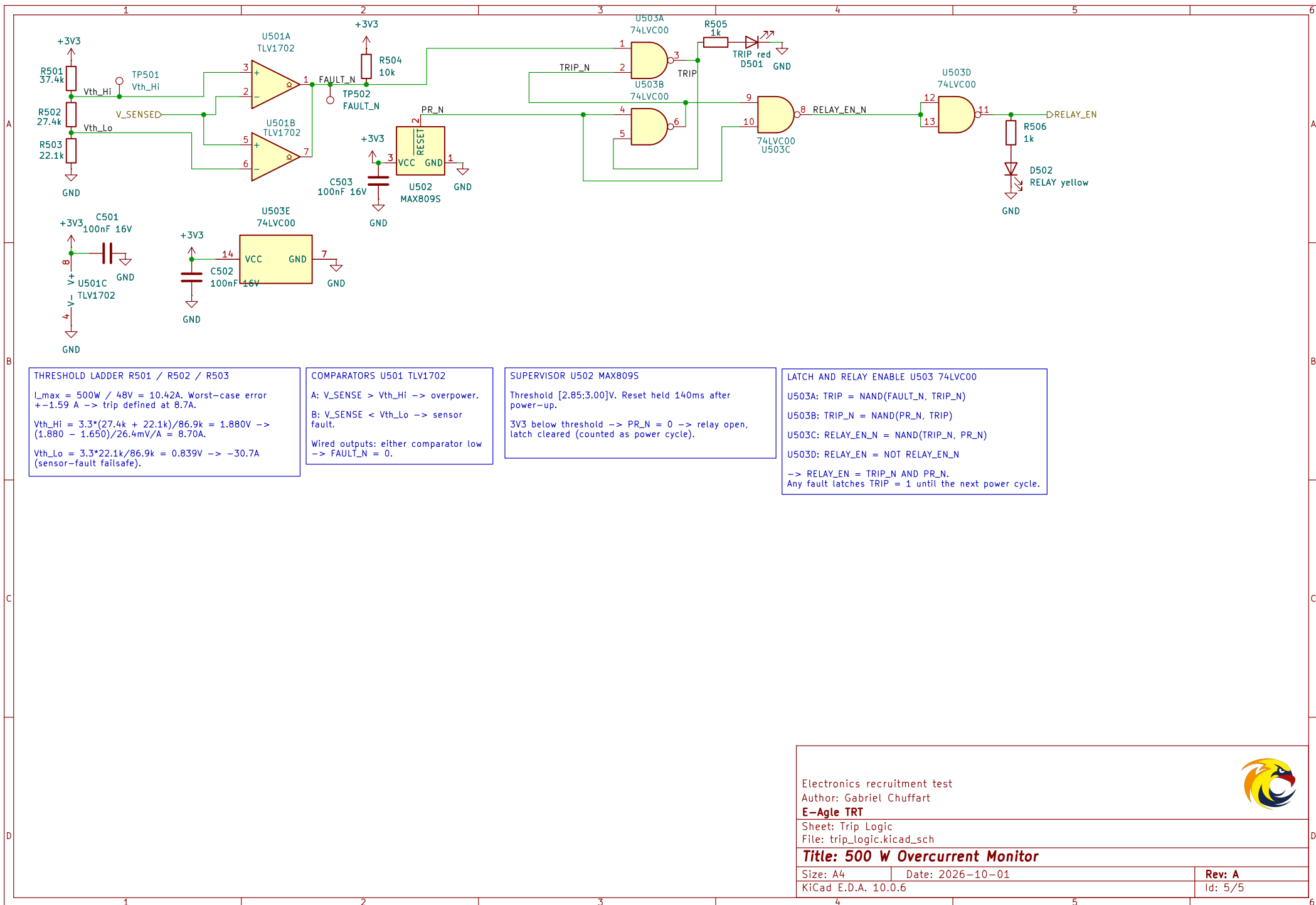
Title: 500 W Overcurrent Monitor

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Rev: A
 Id: 4/5





Electronics recruitment test
Author: Gabriel Chuffart
E-Agle TRT

Sheet: Trip Logic
File: trip_logic.kicad_sch

Title: 500 W Overcurrent Monitor

Size: A4
Date: 2026-10-01
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Rev: A
Id: 5/5

